



SARVAJANIK COLLEGE OF ENGINEERING AND TECHNOLOGY
COMPUTER ENGINEERING DEPARTMENT



B. Tech. IV, CO, SEM - VIII, DIV - I/II
(Term : EVEN - 2025 - 26)

Internship Presentation
on

**“AI-Driven System Engineering: Deployment,
Forecasting, and Document Intelligence”**

Subject Name : Internship (BTCO16804)

Prepared and Presented by
Kuldeep Kevat (Enrollment No. : ET22BTCO053)

Guided by
Prof. Vandana Joshi
Assistant Professor

Details of Company/Industry

- Name : Responscity Systems (P.) Ltd.
- Address : Gujarat Hira Bourse, Surat
- Web-site : <https://responscity.co.in/>
- Industry mentor's Details :
 - Name : Amit Sharma
 - Contact number : 84699 22775
 - E-mail : hello@responscity.com

Presentation Outline

- Face Recognition Optimization
- Water Level Forecasting Models
- OCR & RAG Research
- Complaint AI System Deployment
- Diamond Recommendation System
- Vector-less RAG for DRHP Summarization
- Technical Challenges & Learning

Face Recognition System

Purpose:

- Automate visitor verification system.
- If visitor already registered → face scan → automatic entry.
- If first-time visitor → collect meeting details via UI.
- Reduce manual registration workload.

Existing System:

- UI and backend already developed.
- Face detection & classification implemented.
- Issue: Image retrieval and caching inefficiency.

Face Recognition System

Cache Issues in Existing System

- Cache created on every request/run.
- Cache supported only a single folder.
- No automatic update when:
 - Image added
 - Image deleted
 - Image modified
- Cache not synchronized with database.
- Slow retrieval for large datasets.

Impact

- Response time in minutes.
- Unnecessary cache recreation.
- Incorrect recognition due to outdated cache.
- Poor scalability.

Face Recognition System

Improvements Implemented

- Designed hierarchical folder-level cache system.
- Auto-detects nested folder structure from database.
- Generates cache for all images across multiple levels.
- Cache created:
 - Once at application startup.
 - Not on every request.

Smart Update Logic

- Auto-update cache on:
- Image addition
- Image update
- Image deletion
- No manual GET/PUT request required.
- If full database changes → rebuild cache only at startup.

Results

- Cache creation time reduced from minutes to few seconds.
- Retrieval became near real-time.
- Eliminated redundant cache creation.
- Ensured database-cache consistency.
- Improved system scalability and stability.

Water Level Forecasting Model(Phase 1)

System Modules:

Data Collection

Feature Selection (Discharge + Tidal Height)

LSTM Model Training

Evaluation

Forecast Output

Objective:

Predict next 6 hours water level

Model:

LSTM (Initial Phase)

Used discharge & tidal height

Performance:

92–93% R^2 Score

<15% MAE

5–6% percentage error

Forecast at 10-min intervals

Water Level Forecasting Model(Phase 2)

Objective:

Predict second bridge water level using first bridge trends

Method:

Used previous 6-hour trend

Engineered time-based features

Applied XGBoost Regression

Outcome:

Captured upstream dependency

Accurate short-term prediction

Improved model stability

OCR & RAG Research

OCR Research:

Studied document text extraction

Evaluated structured & unstructured document handling

RAG Study:

Embedding generation

Retrieval mechanisms

Hallucination reduction

Learning:

Context chunking strategies

Retrieval optimization

Context-window handling

Diamond Recommendation System

Logic:

Fetch user preferences

Filter diamonds (cut, clarity, carat, price)

Ranking mechanism

Return top matches

Outcome:

Personalized recommendations

Optimized DB queries

Vector-less RAG for DRHP Summarization(Ongoing)

Objective:

Convert 400–500 page DRHP → 12–15 page structured summary

Pipeline:

File Upload

Company Info Extraction

Industry Classification

Financial Keyword Detection

Table & Metric Extraction

Section Segmentation

Relevant Filtering

LLM Summarization

PDF Generation

Focus:

More tables

Less descriptive text

Financial data prioritization

Conclusion

Successfully deployed production AI system

Built multi-stage forecasting models

Researched advanced RAG architectures

Developed recommendation engine

Designing scalable financial document intelligence system

Internship contributed to:

Real-world AI system deployment

Research-oriented development

Practical ML model implementation

References

- https://www.researchgate.net/publication/360620085_OPTICAL_CHARACTER_RECOGNITION
- <https://arxiv.org/abs/2312.10997>
- https://xgboost.readthedocs.io/en/release_3.2.0/prediction.html
- <http://www.gsdma.org/rainfalldata-2?Type=2>
- <https://pageindex.ai/blog/pageindex-intro>
- <https://www.geeksforgeeks.org/deep-learning/deep-learning-introduction-to-long-short-term-memory/>